Integrated Station Placement and Dispatch Optimisation for Emergency Medical Services: A Case Study on 2.2 Million Seattle 911 Records

Aadith Krishna *CB.SC.U4AIE23202*
Aditya Santosh *CB.SC.U4AIE23207*
Suryansh Ram Menon *CB.SC.U4AIE23255*
Arjungopal Anilkumar *CB.SC.U4AIE23271*

***Abstract*—Emergency medical response time is governed by two decisions that are usually optimised in isolation: where ambulances wait, and which unit is sent to which call. We present an integrated decision-support system that solves both on the same real data and links them, so that the stations opened by the location model become the ambulance bases used by the dispatch model. Using 2.2 million publicly released Seattle Fire 911 records, we filter 79,152 genuine EMS incidents over a twelve-month study period and aggregate them into 28 material demand zones. Three operations-research models are composed in sequence: an Analytic Hierarchy Process (AHP) that ranks candidate sites against incident density, equity gap and coverage potential; set-covering and maximal-covering integer programs that choose station locations; and a transportation-balanced assignment linear program that routes units during a peak hour. On the study period the covering model attains full coverage of all material demand with seven stations at an eight-minute limit, against the thirty-six facilities Seattle operates. Because response times are modelled rather than observed, we re-solve across a grid of travel assumptions and report the resulting range, seven to seventeen stations, rather than the point estimate; coverage remains complete throughout. The assignment LP reduces total fleet travel time by 8.1 % versus nearest-unit-first greedy dispatch at one unit per station, and by 8.8 % at the five-unit fleet that can field the whole peak hour, when both methods are given an identical workload. We show that the naïve version of that comparison overstates the gain roughly sevenfold, and report the corrected figure. Every model is verified against an independent optimum — brute-force enumeration for the covering programs, the Hungarian algorithm for the assignment program — and against randomised instances. The system is delivered as an interactive Streamlit application in which every assumption is adjustable and every model re-solves in under a second.**

***Index Terms*—emergency medical services, facility location, set covering, maximal covering, assignment problem, analytic hierarchy process, integer programming, decision support systems**

I. Introduction

In a cardiac arrest, survival falls by roughly ten percentage points for every minute that passes before defibrillation. Emergency medical services (EMS) are therefore among the oldest and most studied applications of operations research, and the domain in which a better model translates most directly into lives.

Two operational decisions control how quickly an ambulance reaches a patient, and both are optimisable:

- **Q1 — Where should ambulances wait?** Station placement determines how much of the city lies within a given response-time radius.
- **Q2 — Which unit should be sent?** When several emergencies are live at once, the matching of units to calls determines everyone's wait.

These are normally treated as separate literatures: facility location on one side, dispatch and relocation on the other. Treating them separately is convenient but incomplete, because the quality of a dispatch policy is bounded by the geography it inherits. A dispatcher cannot route around a badly placed fleet.

A. Contribution

This paper makes four contributions.

1) **An integrated pipeline.** The station set chosen by the covering model is used directly as the ambulance-base set for the dispatch model, so the two optimisations are evaluated as one system rather than two disconnected exercises (Fig. 1).
2) **A study on real data at full scale.** All 2.2 million records of the Seattle Fire 911 feed are ingested, cleaned and classified, rather than a sampled or synthetic instance.
3) **An honest dispatch benchmark.** We show that comparing an assignment LP against greedy dispatch without controlling the workload measures triage rather than routing, and inflates the apparent benefit from 8.1 % to 62 %. We report the controlled comparison.
4) **A verified, interactive implementation.** Every model is cross-checked against an independent optimum, and the whole pipeline is exposed through a web application in which the travel-time assumptions, study period and response-time limit are adjustable.

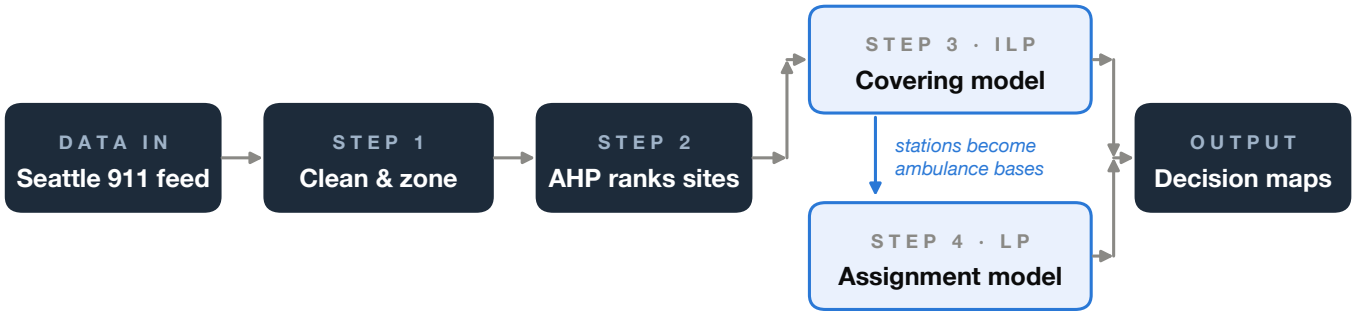


Fig. 1. The integrated pipeline. The two optimisation models are not run side by side: the stations opened by the covering ILP (Step 3) become the ambulance bases of the assignment LP (Step 4), which is the link that makes the system a single decision rather than two.

II. Related Work

A. Covering location models

The location set-covering problem was introduced by Toregas *et al.* [1] precisely for emergency facilities: minimise the number of facilities such that every demand point lies within a critical distance. Because full coverage is often unaffordable, Church and ReVelle [2] proposed the maximal covering location problem (MCLP), which fixes the facility budget at P and maximises covered demand. Both are used here, as complementary readings of the same question. Daskin [3] later extended the family to expected coverage under unit unavailability, and Brotcorne *et al.* [4] survey the ambulance location and relocation literature that grew from these roots. Ahmadi-Javid *et al.* [5] give a broader survey of healthcare facility location, and Başar *et al.* [6] a taxonomy of emergency station location specifically.

B. Dispatch and assignment

The assignment problem has been solvable in polynomial time since Kuhn's Hungarian method [7]. Its linear-programming relaxation is integral because the constraint matrix is totally unimodular [8] — a property we exploit to state dispatch as an LP rather than a binary program, and which we use as a correctness check. Gendreau *et al.* [9] address the dynamic relocation variant with tabu search.

C. Multi-criteria site selection

Saaty's Analytic Hierarchy Process [10] derives priority weights from pairwise comparisons and supplies a consistency ratio that detects self-contradictory judgement. We use AHP not as a competitor to the optimisation models but as the mechanism that converts qualitative stakeholder preference into the candidate ordering the ILP consumes.

D. Response-time thresholds and travel estimation

The eight-minute advanced-life-support benchmark derives from NFPA 1710 [11]; its clinical value is contested [12], [13], which is why our implementation treats the limit as a parameter and reports a sweep rather than a single number. Straight-line distance understates road distance by a roughly constant factor; Boscoe *et al.* [14] measure this circuity nationwide and motivate the 1.3 multiplier used here.

III. Data

A. Sources

Three public, open-licence sources are used; no API keys or accounts are required, and all three are fetched by a script in the repository. The demand side is the Seattle Real-Time Fire 911 Calls feed [17].

TABLE I
Data sources

Source	Content	Role
Seattle Real-Time Fire 911 Calls (`kzjm-xkqj`)	2.2 M incident records	Demand
Seattle Fire Stations (GIS)	36 facilities	Benchmark
US Census TIGERweb ZCTA	ZIP polygons	Zoning

B. Fields used

Each record carries an address with latitude and longitude, a free-text `type` describing the call, a `datetime`, and an `incident_number`. A ZIP zone is computed per incident by point-in-polygon join against the ZCTA layer.

C. Cleaning and EMS classification

The raw feed interleaves fire suppression, alarms, hazardous-materials and medical calls in a single `type` column. Records are de-duplicated on incident number, bounded to the Seattle envelope ($47.40°$–$47.80°$ N, $-122.50°$–$-122.15°$ E), required to carry coordinates, and tagged with a ZIP zone. Of 2,200,120 raw rows, 2,197,739 survive cleaning.

EMS calls are then isolated by whole-word pattern matching on the call type (*aid*, *medic*, *medical*, *AMR*, *nurseline*, *triaged*, *low acuity*, *overdose*, and similar). A second, opt-in tier adds incidents that generate patients but are typed by their cause — collisions, rescues, violence — so that the sensitivity of every result to the definition of "EMS" can be inspected rather than assumed. All results in this paper use the core tier.

D. Study period and demand zones

The feed is harvested live and spans 2003 to the present; its final calendar day is truncated mid-harvest and is flagged rather than silently dropped. We take the most recent twelve months, 27 Aug 2025 to 27 Aug 2026, giving **79,152 EMS incidents**, an average of 216 per day.

Each ZIP zone becomes a demand point whose weight is its incident count and whose location is the *mean incident coordinate* rather than the polygon centroid — placing the candidate site where the calls actually are.

E. The materiality floor

The feed carries a long tail of ZIP zones holding a handful of mutual-aid calls per year in suburbs outside the city (Renton, SeaTac, Medina and similar). Set covering treats a one-call zone exactly like a ten-thousand-call zone, so if these are left in, each demands its own dedicated station: the eight-minute answer inflates from 7 stations to 13, six of them serving 0.04 % of demand.

A materiality floor (default 0.1 % of period demand, adjustable) therefore excludes such zones from the models while keeping them visible in every table. Of 39 ZIP zones in the period, **28 are material**; the 11 excluded hold 161 incidents in total, so the floor retains **99.80 %** of all demand. This is a modelling decision with a visible effect on the headline result, and is reported as such rather than buried in preprocessing.

F. Demand structure

Fig. 2 shows that demand is strongly concentrated: the busiest zone, ZIP 98104, alone carries 11.9 % of all EMS incidents, and 16 of 39 zones account for 80 % of them. Fig. 3 shows the temporal profile — a pronounced midday peak at 13:00 and a shallow weekday pattern. Demand is therefore far more concentrated in space than in time, which is what makes station placement the higher-leverage of the two decisions.

IV. Methodology

A. Travel-time model

Great-circle distance between two points is computed by the haversine formula. Real ambulances follow streets, so the straight-line distance is inflated by a *circuity factor* κ, converted at an *effective response speed* v, and a *turnout time* τ is added, so that a threshold means time from alarm to on-scene:

$$t_{ij} \;=\; \tau \;+\; \frac{60\,\kappa\, d_{ij}}{v} \tag{1}$$

where d_{ij} is the haversine distance in kilometres and t_{ij} is in minutes. Defaults are $\kappa = 1.30$ [14], $v = 45$ km/h and $\tau = 1$ min. All three are exposed in the application and every model re-solves against the current values, so no result in this paper depends on an unexaminable constant.

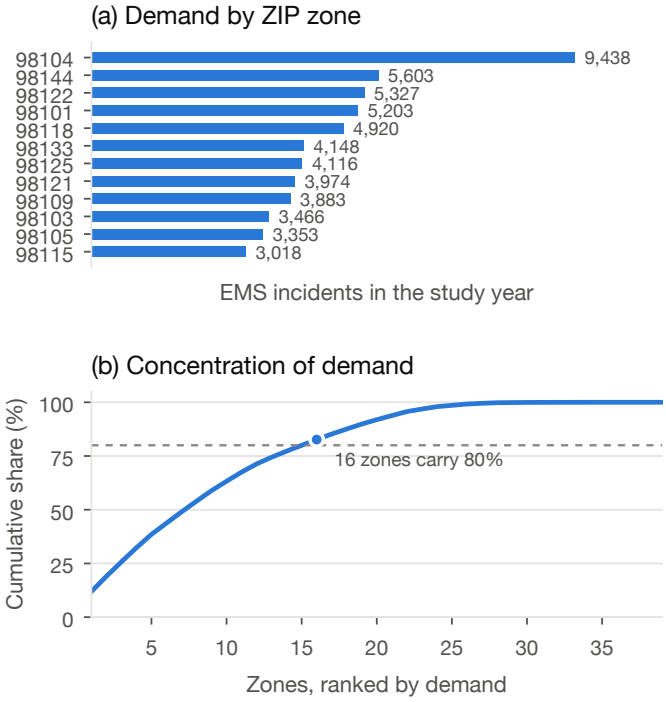


Fig. 2. Spatial concentration of EMS demand over the study year.

B. Step 2 — AHP site scoring

Three competing benefit criteria are defined per candidate zone: *Incident Density* (calls in the zone), *Equity Gap* (kilometres to the nearest station Seattle already operates — farther means worse served), and *Coverage Potential* (demand reachable from the zone within the limit).

Each of the four team members independently compared the three criteria pairwise on the Saaty 1–9 scale, giving three judgements each. Member k's reciprocal matrix $A^{(k)}$ is built from those, and the four matrices are combined by the element-wise geometric mean,

$$\bar{a}_{ij} \;=\; \left(\prod_{k=1}^{4} a_{ij}^{(k)}\right)^{1/4}, \tag{2}$$

which is the only average that preserves reciprocity: if the members' views of i versus j average to 3, their view of j versus i automatically averages to $1/3$. The priority vector w is the principal eigenvector of $\bar{A}$, and consistency is checked by

$$\mathrm{CI} = \frac{\lambda_{\max} - n}{n-1}, \qquad \mathrm{CR} = \frac{\mathrm{CI}}{\mathrm{RI}_n}, \tag{3}$$

with $\mathrm{RI}_3 = 0.58$ and the usual acceptance rule $\mathrm{CR} < 0.1$.

Each zone's criterion values are max-normalised — appropriate because all three are benefit criteria measured on ratio scales with a true zero — and combined into a priority score $s_i = \sum_c w_c\, \hat{x}_{ic}$. This score orders the candidate list the covering model receives, and additionally enters the set-covering objective as an infinitesimal tie-break, so that among

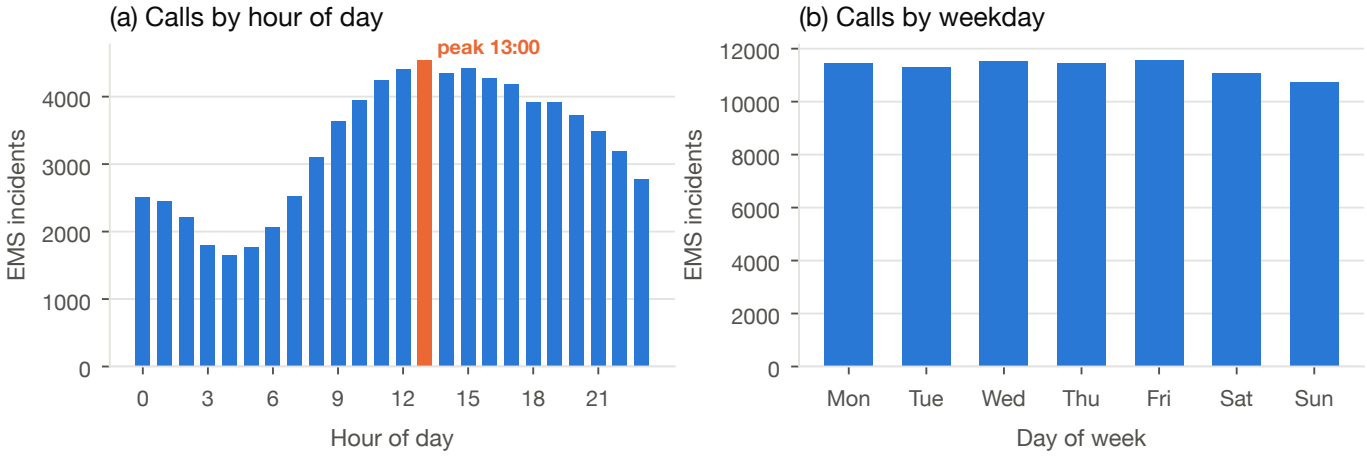


Fig. 3. Temporal profile of EMS demand. The midday peak is pronounced; the weekday variation is not. The busiest single hour in the period, used for the dispatch scenario, is 09:00 on Wednesday 19 November 2025, with 30 simultaneous incidents.

equally sized optimal solutions the model prefers the one on higher-priority sites.

C. Step 3 — Covering models (ILP)

Let I be candidate sites, J demand zones, d_j the demand of zone j, and $N(j) = \{\, i \in I : t_{ij} \le T \,\}$ the sites that can reach j within limit T. Let $y_i \in \{0,1\}$ open site i.

Mode A — Set covering. The fewest stations that leave no zone uncovered:

$$\begin{aligned} \min_{y} \quad & \sum_{i \in I} y_i && && (4)\\ \text{s.t.} \quad & \sum_{i \in N(j)} y_i \;\ge\; 1 && \forall j \in J \\ & y_i \in \{0,1\} && \forall i \in I \end{aligned}$$

Mode B — Maximal covering. With the budget fixed at P, the greatest demand served, where $z_j \in \{0,1\}$ indicates that zone j is covered:

$$\begin{aligned} \max_{y,z} \quad & \sum_{j \in J} d_j z_j && && (5)\\ \text{s.t.} \quad & z_j \;\le\; \sum_{i \in N(j)} y_i && \forall j \in J \\ & \sum_{i \in I} y_i = P \\ & y_i,\, z_j \in \{0,1\} \end{aligned}$$

Zones that no candidate can reach within T are structurally impossible to cover; they are reported separately and excluded from the constraint set, so that an unreachable outlier cannot render the whole program infeasible and suppress an otherwise useful answer.

D. Step 4 — Dispatch assignment (LP)

For a chosen peak hour, let c_{ij} be the travel time from ambulance i to incident j under (1), and x_{ij} the fraction of unit i assigned to call j:

$$\begin{aligned} \min_{x} \quad & \sum_{i} \sum_{j} c_{ij} x_{ij} && && (6)\\ \text{s.t.} \quad & \sum_{i} x_{ij} = 1 && \forall j && (7)\\ & \sum_{j} x_{ij} \le 1 && \forall i && (8)\\ & x_{ij} \ge 0 \end{aligned}$$

The constraint matrix of (6) is totally unimodular [8], so the LP relaxation has an integral optimum and no binary variables are needed — which is why dispatch is stated here as an LP while covering requires an ILP. Unequal unit and call counts are handled in the transportation-problem manner, by padding the cost matrix with zero-cost dummy rows or columns so that the instance is square, balanced and always feasible.

E. The integrating link

The set $\{\, i : y_i = 1 \,\}$ returned by (4) is passed directly as the ambulance-base set of (6). A `units_per_station` parameter replicates each base so that an hour with more calls than stations can still be served. This is the link that makes the two models one system, and it is the principal methodological contribution of the work.

F. Why the obvious dispatch benchmark is wrong

A natural way to show the value of optimal dispatch is to compare the LP objective against nearest-unit-first greedy assignment on the same peak hour. This comparison is invalid, and we report it because the invalid version is the one most likely to be produced by default.

A peak hour carries more calls than ready units — 30 calls against 7 units in our instance. Under (6) an unserved call

contributes nothing to the objective. The LP is therefore free to answer only the seven *nearest* calls and leave the awkward ones unanswered, while greedy processes calls in arrival order and must take whatever comes. Comparing the two totals then scores *which calls each method chose to answer* — that is, triage — rather than how well it routed, and inflates the apparent advantage from 8.1 % to 62 %.

We therefore restrict the benchmark to an identical workload on both sides: the first k calls by arrival time, where k is the fleet size. Both methods then clear exactly the same calls, and the difference measures matching quality alone. All improvement figures reported in this paper are of this controlled form.

V. System Implementation

The system is a Streamlit web application of roughly 4,000 lines of Python, organised so that every model is a pure, independently testable module and the interface is a thin presentation layer over them.

TABLE II
Repository structure

Module	Responsibility
`src/config.py`	Paths, EMS taxonomy, constants, AHP judgements
`src/geo.py`	Haversine, travel time, point-in-polygon join
`src/data_loader.py`	Ingestion, cleaning, zones, peak hours
`src/stats.py`	On-demand descriptive statistics
`src/ahp.py`	Pairwise matrices, weights, consistency
`src/covering.py`	Set covering & maximal covering ILPs
`src/assignment.py`	Dispatch assignment LP, greedy baseline
`src/viz.py`	Plotly charts, Folium maps, colour tokens
`app.py`	Four-tab Streamlit interface

A. Interface

The application presents four tabs mirroring the pipeline of Fig. 1: **Data** (demand distribution, temporal profile, the weights entering both optimisations), **Site scoring** (a per-member AHP comparison tab, the aggregate matrix and the consistency check), **Station placement** (both covering modes, the threshold sweep, and the benchmark against the real network) and **Dispatch** (the assignment result and the greedy comparison). A sidebar exposes the study period, the EMS tier, the materiality floor and all three travel-time parameters; changing any of them re-solves every downstream model.

B. Solver portability

The models are expressed with PuLP [15]. PuLP bundles a CBC binary built for x86_64 only, so on Apple Silicon without Rosetta every solve fails with *"Bad CPU type in executable"* — and PuLP reports the broken binary as available, because only the exec fails. The configuration layer therefore probes candidate solvers by *actually solving a trivial LP* and uses the first that runs: HiGHS [16], then a system CBC on `PATH`, then PuLP's bundled copy. All results here were produced with HiGHS.

C. Performance

The cleaned incident table is cached to Parquet and loaded once per session. Two representation choices halve its footprint from 332 MB to 173 MB: a redundant `date` column duplicating `datetime` is excluded at read time, and the high-cardinality `type` and `zip_code` columns are loaded as categoricals. Neither changes any result. With the table resident, every model in this paper solves in well under a second (Table III).

TABLE III
Solve times on the study instance (HiGHS, Apple Silicon)

Model	Size	Time (s)
Set covering, $T = 8$	28 sites × 28 zones	0.005
Assignment, 7 units	30×30 balanced	0.013
Assignment, 14 units	30×30 balanced	0.014

VI. Results

A. AHP weights and consistency

Table IV gives the four members' judgements and the resulting weights. Every member's matrix is perfectly consistent ($\mathrm{CR} = 0.000$), as is the aggregate. This is a property of the ratings rather than good fortune: each member's density-versus-coverage rating is exactly the product of their other two judgements, so the matrices contain no contradiction for the consistency check to find. We note this explicitly because a uniformly zero CR invites the suspicion that the check is not running; it is, and §VII describes the randomised test that confirms it.

TABLE IV
Team AHP judgements and derived weights

	Saaty ratings			Weights (%)			
Member	D:E	D:C	E:C	D	E	C	**CR**
Aadith Krishna	3	6	2	66.7	22.2	11.1	0.000
Aditya Santosh	2	6	3	60.0	30.0	10.0	0.000
Suryansh Ram Menon	4	8	2	72.7	18.2	9.1	0.000
Arjungopal Anilkumar	3	9	3	69.2	23.1	7.7	0.000
Aggregate	2.91	7.14	2.45	**67.4**	**23.1**	**9.4**	0.000

D = Incident Density, E = Equity Gap, C = Coverage Potential. Aggregate ratings are the element-wise geometric mean, (2). $\lambda_{\max} = 3.000$, $\mathrm{CI} = 0.000$, $\mathrm{RI}_3 = 0.58$.

The team weights raw call volume (67.4 %) far above distance from existing help (23.1 %), and rates the ability to blanket several zones from one site (9.4 %) least important of the three. The top-ranked candidate sites under these weights are ZIP 98104 (score 0.778), 98144 (0.579), 98101 (0.534) and 98122 (0.512).

B. Station placement

Table V and Fig. 4 give the central result. At an eight-minute limit, **seven stations cover 100 % of material demand**; tightening to five minutes requires sixteen, and relaxing to ten minutes needs only five. Section VI-C qualifies the seven: it is the count under the default travel constants, and the figure moves substantially under others. Fig. 5 shows the eight-minute solution against the thirty-six stations Seattle actually operates.

TABLE V
COVERAGE ATTAINED, OPTIMISED PLACEMENT VS. THE REAL NETWORK

Limit	Optimised		Seattle's real network	
	Stations	Demand	Stations	Demand
5 min	16	100.00 %	36	99.72 %
8 min	7	100.00 %	36	100.00 %
10 min	5	100.00 %	36	100.00 %

Fig. 7 gives the maximal-covering curve that justifies the budget-constrained framing: the first station alone covers 51 % of demand, five cover 98 %, and the last two buy the remaining 1.8 %. Returns diminish sharply, which is exactly the regime in which MCLP rather than set covering is the right question to ask.

C. Robustness of the station count

The seven-station figure is a statement about two constants as much as about Seattle. The 911 feed records no response times, so every travel time in this study is computed from (1): a straight-line distance inflated by a circuity factor and divided by an effective speed. Both are chosen, not measured, and the coverage radius they imply scales linearly in one and inversely in the other.

We therefore re-solve the set-covering program over a 4×4 grid of those constants, holding the candidate pool and the AHP tie-break scores fixed so the spread is attributable to the travel model alone. Speeds range over 30 to 45 km/h and circuity over 1.3 to 1.6, spanning optimistic and pessimistic readings of the same street network. Each cell is an exact ILP solve; Fig. 6 reports the sixteen.

The station count required for full coverage ranges from **seven to seventeen**, a factor of 2.4, while demand covered remains at 100 % in every cell. The two readings of that are different in kind. The *count* is assumption-dependent and should not be quoted alone; the *comparison* against the 36-site real network is not, since seventeen is still less than half of thirty-six. We report the range for this reason, and treat the directional claim rather than the point estimate as the result.

The grid is monotone in both directions, which is a necessary condition for it to be measuring what it claims: a slower fleet or a more circuitous network can never require fewer stations. This is asserted in the verification suite, and distinguishes a genuine gradient from solver noise.

D. Dispatch

The busiest hour in the study period is 09:00 on Wednesday 19 November 2025, with 30 simultaneous incidents. Table VI gives the controlled comparison of §IV-F, also shown in Fig. 8.

TABLE VI
OPTIMAL VS. GREEDY DISPATCH, LIKE-FOR-LIKE WORKLOAD

Fleet	Calls	Optimal	Greedy	Gain
7 units (1/station)	7	74.40 min	81.00 min	**8.1 %**
14 units (2/station)	14	143.96 min	149.91 min	**4.0 %**
35 units (5/station)	30	252.06 min	276.38 min	**8.8 %**

The gain is not monotone in fleet size: 8.1 %, 4.0 % and 8.8 % at seven, fourteen and thirty-five units. Each row is scored on a different workload — the first $k = \min(\text{units, calls})$ arrivals — so the three are not points on a curve but three separate instances, and which particular calls fall inside k matters more than the fleet size itself. We therefore report the margin per configuration rather than as a single figure, and draw no trend from three instances.

VII. VERIFICATION

Because every number above is produced by a solver rather than derived in closed form, the implementation is verified by two automated suites totalling 90 checks, both of which pass in full.

The first suite asserts the *properties* each model must satisfy on the real data: set covering must reach full coverage of every reachable zone; maximal covering must open exactly P and be non-decreasing in P; MCLP at P equal to the set-cover optimum must attain the same coverage; a looser limit must never require more stations; AHP weights must sum to one with $\lambda_{\max} \geq n$ and $\mathrm{CR} < 0.1$; the greedy benchmark must place the same workload on both sides; and the LP must never lose to greedy.

The second establishes *optimality* against independent oracles:

- both covering models are re-solved on small synthetic cities and checked against exhaustive enumeration of every possible station subset (25 instances each);
- the AHP weights are checked against the largest-modulus eigenvector computed independently, over 200 random reciprocal matrices — the test that confirms the consistency machinery is live despite the zero CRs in Table IV;
- the dispatch LP is cross-checked against an independent Hungarian algorithm solution of the same balanced matrix, and against greedy over 150 random cost matrices.

A third suite drives the application headlessly and asserts that every pipeline stage has its tab, every headline figure is on the page, and all three maps build — so the parts of the deliverable that are looked at rather than computed cannot silently regress.

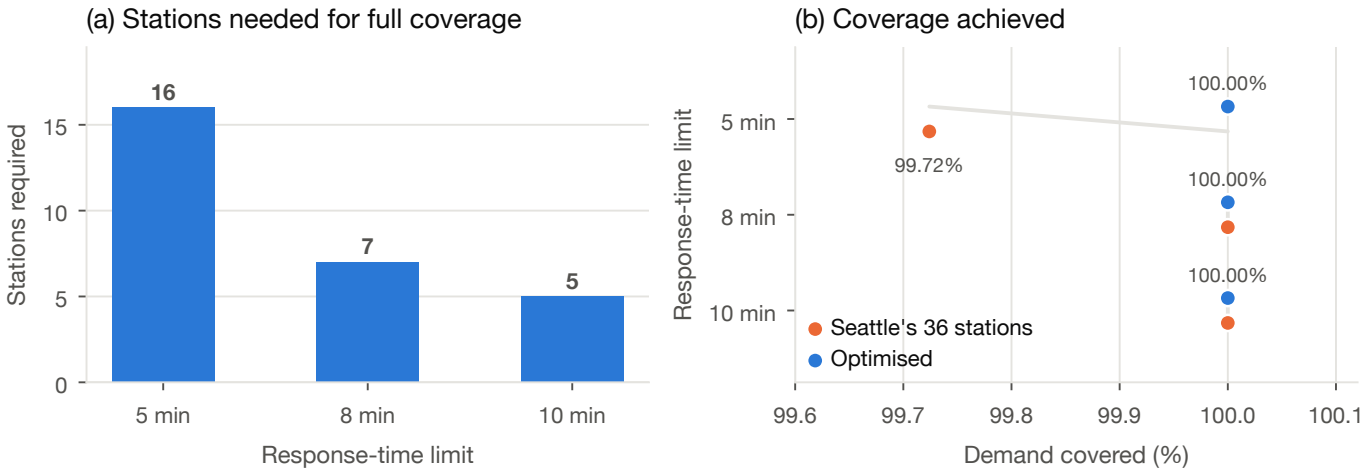


Fig. 4. Sensitivity to the response-time limit. Panel (b) uses a dot plot rather than bars because the two networks differ only in the second decimal; a truncated bar baseline would misrepresent a 0.28-point gap as a rout.

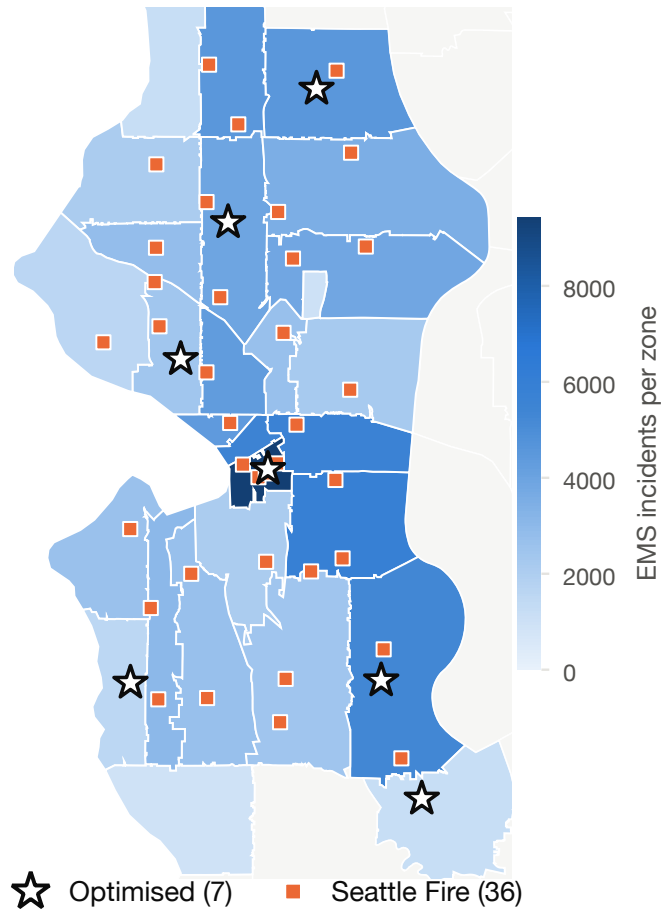


Fig. 5. The seven-station eight-minute solution (stars) against Seattle's 36 operating fire stations (squares), over EMS demand by ZIP zone.

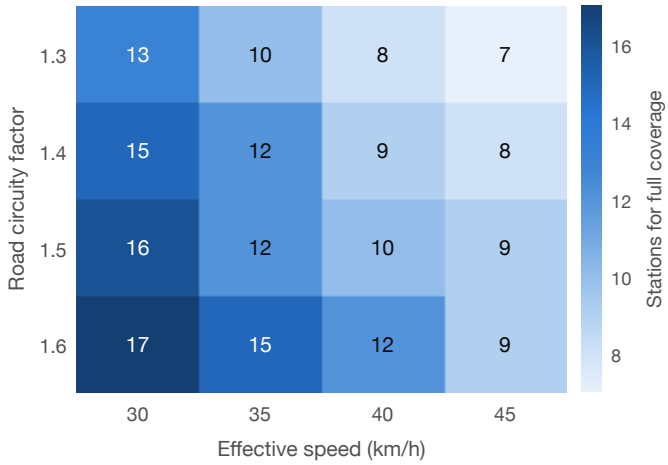


Fig. 6. Stations required for full coverage at an eight-minute limit, over a grid of the two travel constants. The count varies from seven to seventeen; coverage is complete in all sixteen cells.

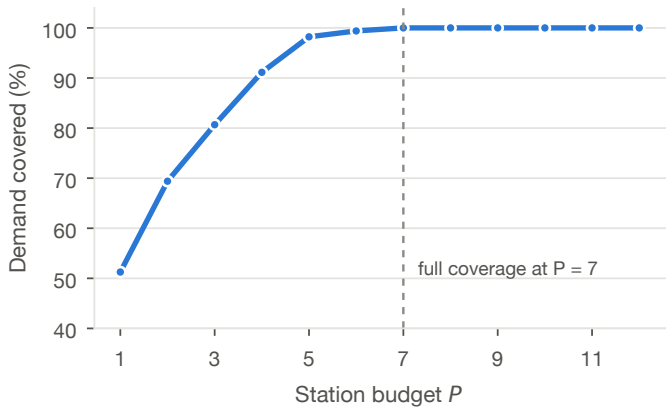


Fig. 7. Marginal coverage against station budget P at an eight-minute limit. Five stations already reach 98.2 % of demand.

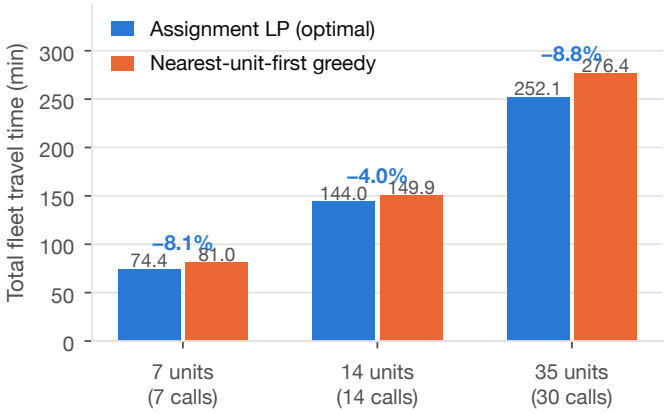


Fig. 8. Total fleet travel time, optimal assignment against nearest-unit-first greedy dispatch, with both methods clearing an identical set of calls.

VIII. Discussion

A. Why the optimised network needs fewer sites

The optimised network reaches the same coverage from a fraction of the sites. The fraction itself depends on the travel assumptions (Section VI-C), but it is below one in every combination tested, and the reason is not that Seattle Fire is badly run. Three factors explain the gap, and stating them is necessary for the result to be read correctly.

The real network is not solving this problem. The 36 benchmark sites house fire suppression, rescue and hazardous-materials response as well as EMS, and two of them are the department headquarters and Harborview Medical Center rather than numbered stations. Our model prices only EMS demand, and a station sited for structural fire risk in a low-call-volume district is doing work this objective cannot see.

Placement follows demand rather than history. Real station networks accrete over a century of annexations, land availability and politics. An optimiser starting from a blank map has no such inheritance.

Zone-centroid demand is optimistic. Coverage is evaluated at 28 demand-weighted centroids, not at 79,152 individual incidents. A station that covers a centroid does not necessarily reach every call in that zone within the limit. This is the single largest source of optimism in the result, and §IX treats it as such.

The honest reading is therefore not "Seattle should close 29 fire stations". It is that *EMS demand in Seattle is concentrated enough that a small, demand-sited ambulance fleet can cover it*, which is an argument about where to put ambulances, not about how many fire stations a city needs.

B. Spatial concentration is the operative fact

The Pareto curve of Fig. 2(b) and the marginal curve of Fig. 7 are the same fact seen twice. Because 16 of 39 zones carry 80 % of demand, a handful of well-placed sites captures most of the benefit, and the covering models have an easy instance. A city with flatter demand would produce a flatter marginal curve and a correspondingly larger optimal station count.

C. Routing gains are real but modest

The 8.1 % improvement from optimal dispatch is a genuine gain available at zero capital cost — it requires only a different decision rule at the console. But it is an order of magnitude smaller than the apparent 62 % that the uncontrolled comparison reports, and we regard the discipline of §IV-F as among the more transferable results here: in any assignment benchmark where the objective is free to leave work undone, the baseline must be constrained to the same workload or the comparison measures selection rather than skill.

IX. Limitations

We state these explicitly rather than leaving them to be discovered.

1) **Zone-level aggregation.** Demand is modelled at 28 zone centroids. Incident-level coverage would be stricter and would raise the station count.
2) **Euclidean travel with a constant circuity factor.** A single multiplier cannot represent bridges, the ship canal, or congestion that varies by hour. Seattle's geography is unusually hostile to this approximation. A road-network routing engine would be the natural replacement.
3) **Static, deterministic model.** Units are assumed available at their base at the start of the hour. Queueing, busy fractions, hospital turnaround and relocation are not modelled; Daskin's expected-covering family [3] is the standard extension.
4) **One objective per model.** Total travel time is minimised, which does not control the worst case; a minimax or equity-constrained variant would trade mean response for tail response.
5) **AHP judgements are from four student raters**, not EMS practitioners, and all four happened to be perfectly consistent. The weights should be read as a demonstration of the mechanism rather than as elicited domain preference.
6) **A single peak hour** is used for the dispatch scenario. Results across many hours would be a stronger claim than one.

X. Conclusion

We have presented an integrated decision-support system that solves EMS station placement and ambulance dispatch on the same real data and links them, so that the stations chosen by a covering ILP serve as the ambulance bases of a dispatch LP. On 79,152 real Seattle EMS incidents the system attains full coverage of all material demand with seven stations at an eight-minute limit, or seven to seventeen once the modelled travel constants are varied over a plausible grid, and reduces total fleet travel time by 8.1 % to 8.8 % depending on fleet size over nearest-unit-first greedy dispatch under a workload-controlled comparison. Every model is cross-checked against an independent optimum.

Two results generalise beyond this instance. First, the integration itself: evaluating dispatch on an optimised geography rather than an arbitrary one changes what the dispatch numbers mean, and the two models should be reported together. Second, the benchmark discipline of §IV-F: an assignment objective that is free to leave calls unserved will flatter itself against a baseline that is not, and we found the inflation to be roughly sevenfold.

Future work follows the limitations directly: road-network travel times in place of circuity-adjusted haversine, incident-level rather than zone-level coverage, a probabilistic availability model in the Daskin tradition, and evaluation across many peak hours rather than one.

AVAILABILITY

The full implementation, both verification suites and the scripts that download and preprocess the data are available in the project repository. All data sources are public and open-licence.

REFERENCES

[1] C. Toregas, R. Swain, C. ReVelle, and L. Bergman, "The location of emergency service facilities," *Operations Research*, vol. 19, no. 6, pp. 1363–1373, 1971.

[2] R. Church and C. ReVelle, "The maximal covering location problem," *Papers of the Regional Science Association*, vol. 32, no. 1, pp. 101–118, 1974.

[3] M. S. Daskin, "A maximum expected covering location model: formulation, properties and heuristic solution," *Transportation Science*, vol. 17, no. 1, pp. 48–70, 1983.

[4] L. Brotcorne, G. Laporte, and F. Semet, "Ambulance location and relocation models," *European Journal of Operational Research*, vol. 147, no. 3, pp. 451–463, 2003.

[5] A. Ahmadi-Javid, P. Seyedi, and S. S. Syam, "A survey of healthcare facility location," *Computers & Operations Research*, vol. 79, pp. 223–263, 2017.

[6] A. Başar, B. Çatay, and T. Ünlüyurt, "A taxonomy for emergency service station location problem," *Optimization Letters*, vol. 6, no. 6, pp. 1147–1160, 2012.

[7] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1–2, pp. 83–97, 1955.

[8] A. J. Hoffman and J. B. Kruskal, "Integral boundary points of convex polyhedra," in *Linear Inequalities and Related Systems*, H. W. Kuhn and A. W. Tucker, Eds. Princeton, NJ: Princeton University Press, 1956, pp. 223–246.

[9] M. Gendreau, G. Laporte, and F. Semet, "Solving an ambulance location model by tabu search," *Transportation Research Part C*, vol. 5, no. 2, pp. 75–88, 1997.

[10] T. L. Saaty, "A scaling method for priorities in hierarchical structures," *Journal of Mathematical Psychology*, vol. 15, no. 3, pp. 234–281, 1977.

[11] National Fire Protection Association, *NFPA 1710: Standard for the Organization and Deployment of Fire Suppression Operations, Emergency Medical Operations, and Special Operations to the Public by Career Fire Departments*. Quincy, MA: NFPA, 2020.

[12] P. T. Pons and V. J. Markovchick, "Eight minutes or less: does the ambulance response time guideline impact trauma patient outcome?" *The Journal of Emergency Medicine*, vol. 23, no. 1, pp. 43–48, 2002.

[13] T. H. Blackwell and J. S. Kaufman, "Response time effectiveness: comparison of response time and survival in an urban emergency medical services system," *Academic Emergency Medicine*, vol. 9, no. 4, pp. 288–295, 2002.

[14] F. P. Boscoe, K. A. Henry, and M. S. Zdeb, "A nationwide comparison of driving distance versus straight-line distance to hospitals," *The Professional Geographer*, vol. 64, no. 2, pp. 188–196, 2012.

[15] S. Mitchell, M. O'Sullivan, and I. Dunning, "PuLP: a linear programming toolkit for Python," Dept. Eng. Science, Univ. of Auckland, Tech. Rep., 2011.

[16] Q. Huangfu and J. A. J. Hall, "Parallelizing the dual revised simplex method," *Mathematical Programming Computation*, vol. 10, no. 1, pp. 119–142, 2018.

[17] City of Seattle, "Seattle Real-Time Fire 911 Calls," Seattle Open Data Portal, dataset `kzjm-xkqj`. [Online]. Available: https://data.seattle.gov/